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A Cornerstone Project

AI Resume Analyzer

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Presentation Outline

1.Project overview
2.Problem and objectives
3.System architecture
4.Input and preprocessing
5.Skill recognition
6.Vector scoring
7.Scoring example
8.3D skill map
9.Upload screen
10.Matched and missing skills
11.Verification and limitations
12.Conclusion

PROJECT OVERVIEW

Abstract

Matchspace compares named technical skills in an uploaded job description and resume. It embeds each recognized label with a pretrained sentence model, assigns one-to-one matches, and returns a scored explanation with a rotatable 3D view.

Introduction

A user supplies a UTF-8 JD text file and a selectable-text resume PDF. The result shows matched and missing JD skills, point credit for each selected pair, and the overall score. The tool supports document review; it does not assess a person’s competence.

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PROBLEM STATEMENT & OBJECTIVES

Problem statement

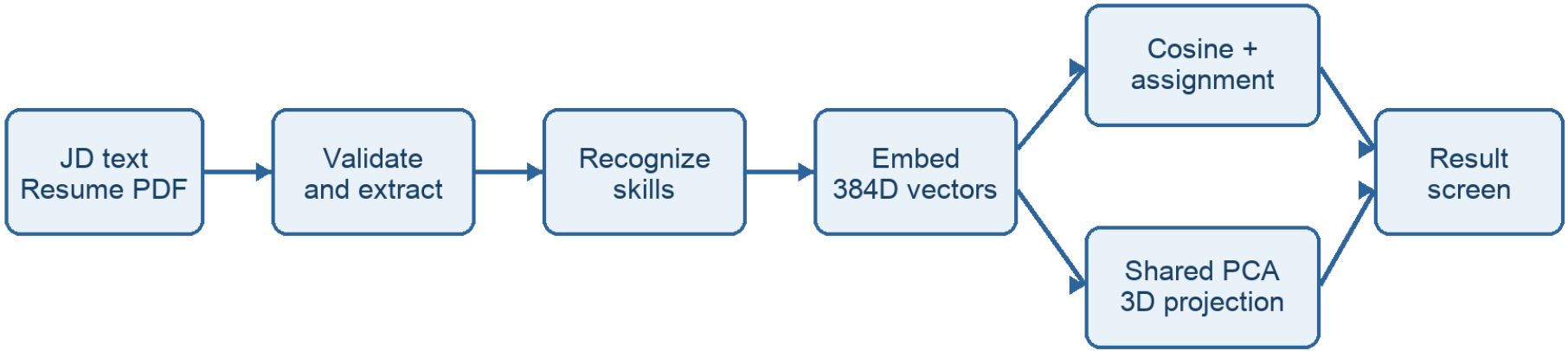
Direct keyword search can miss related terms, while a single percentage hides which requirements were recognized. The project needs a visible, repeatable comparison of technical skills across a JD and a resume.

Objectives

- Accept a JD text file and selectable-text resume PDF.
- Recognize distinct technical skills and known aliases.
- Compare skill embeddings with cosine similarity.
- Assign each resume skill to at most one JD skill.
- Show the score, matched pairs, and missing skills.
- Display both skill sets in a rotatable 3D map.

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SYSTEM ARCHITECTURE



Scoring uses complete vectors

The 3D projection only controls display coordinates

INPUT & PREPROCESSING

Accepted files

The upload screen accepts a UTF-8 .txt job description up to 250 KB and a selectable-text .pdf resume up to 5 MB. The server validates file types, sizes, and readable content before analysis. An image-only PDF receives a clear OCR-related error.

Document focus

When the JD has a Required Qualifications or Requirements section, that section supplies the scored requirements. Otherwise the matcher prefers a named technical-skills section. The resume reader uses PDF layout extraction and skips contact or URL-like lines.

SKILL RECOGNITION

Curated vocabulary

The matcher searches for supported technical names and aliases, including Python, Docker, Kubernetes, CI/CD, Terraform, observability, and SRE terms. It deduplicates repeated mentions under a canonical skill name while keeping a readable label for the result.

Recognition boundary

Known variants such as K8s and Kubernetes count as one skill. Unknown or ambiguous skills may be missed because the vocabulary is finite. The system extracts named terms; it does not infer every skill from unrestricted resume prose.

VECTOR MATCHING

Embedding and similarity

The pretrained all-MiniLM-L6-v2 model converts each recognized skill label into a 384-dimensional vector. The matcher compares JD and resume vectors with cosine similarity. Opaque abbreviations are expanded before encoding, while visible labels remain unchanged.

Points and assignment

Pair credit is clamp((cosine − 0.60) / 0.40, 0, 1). A one-to-one assignment selects the set with the highest total credit, so one resume skill cannot score twice. The percentage is 100 × earned credit / recognized JD-skill count.

SCORING EXAMPLE

Synthetic demonstration

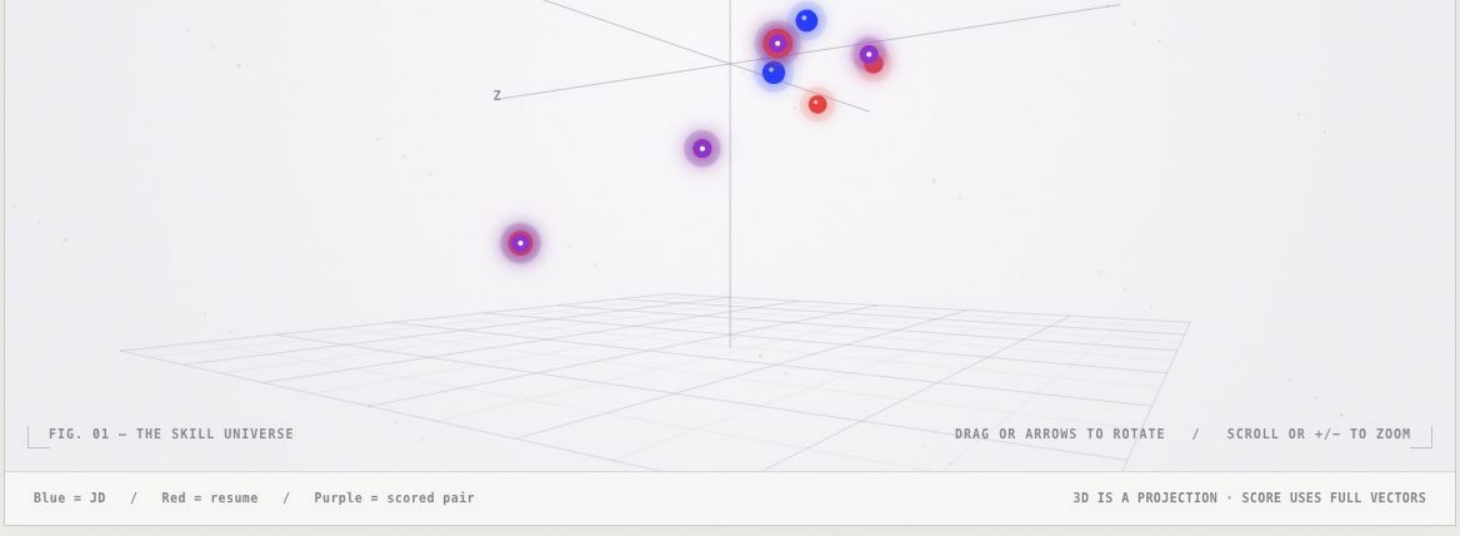
A repeatable test JD lists seven technical skills. The synthetic resume shares five labels and includes two additional skills. The browser run shows five exact pairs, two missing JD skills, and a score of 71.43%.

Observed result

- Python, Bash, Docker, Kubernetes, and CI/CD earn 1.00 point each.
- Terraform and Observability remain unmatched.
- Total credit is 5.00 out of seven possible skill points.
- The displayed percentage is 100 × 5 / 7 = 71.43%.
- Purple markers show the selected scored pairs.
- This is a functional example with synthetic files.

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3D SKILL MAP



THE MATCH SCORE

71.43%

LOOK CLOSER / 02

Where you connect.

Matches 5 Missing 2

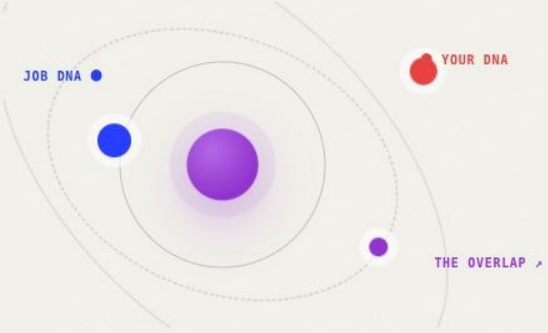
Python ↔ Python	+1.00 PT
Bash ↔ Bash	+1.00 PT

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UPLOAD SCREEN

Where skills find their *orbit.*

Drop in a job description and your resume. Watch the distance between your skills become something you can actually see.



✳ BUILT FOR CURIOUS CAREERS — 2026

START HERE / 01

Bring your files.

01 Job description — TXT ONLY

matchspace_demo_jd.txt
117 B · ready to map

02 Your resume — PDF ONLY

matchspace_demo_resume.pdf
2 KB · ready to map

Map the match

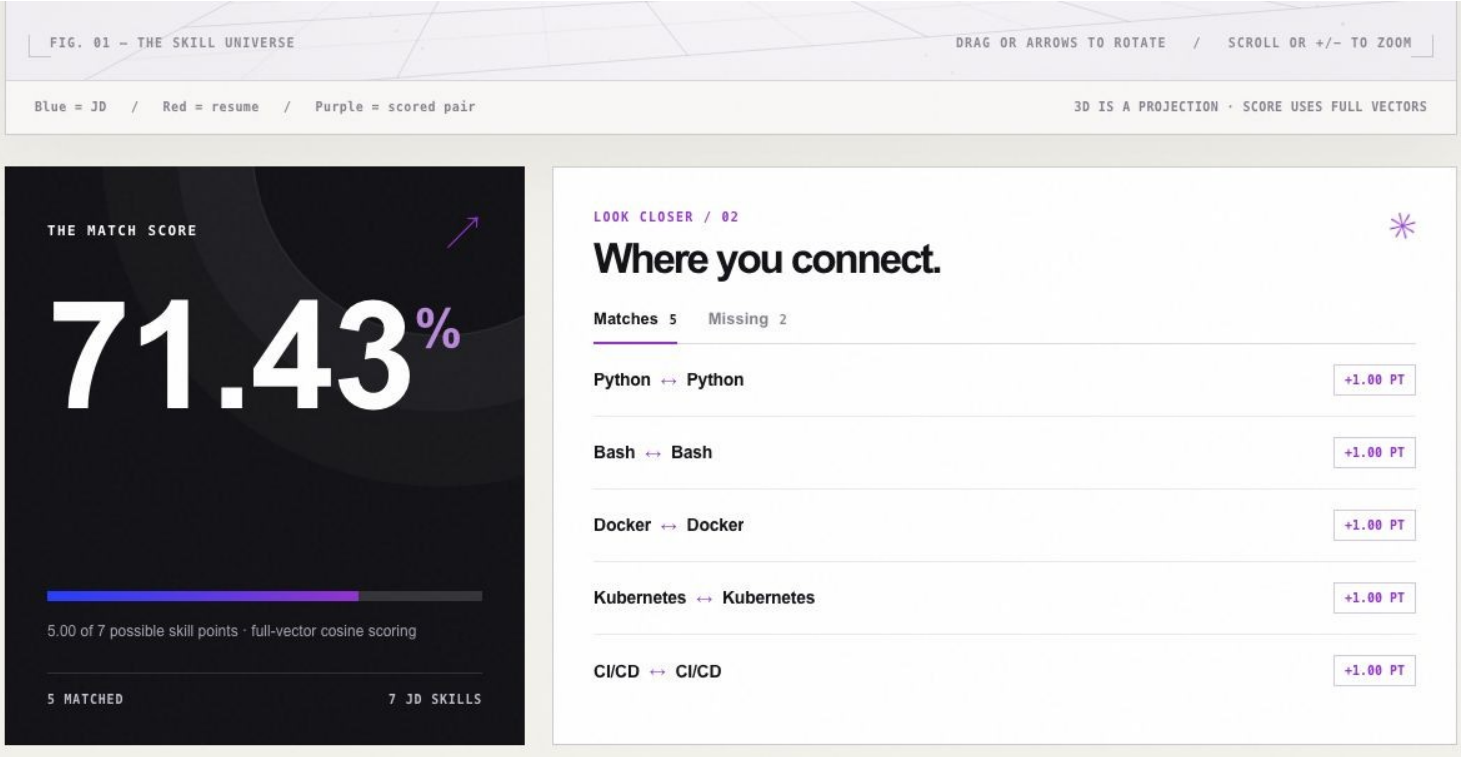
✳ Your files are processed locally and are not saved by the app.

01 / UPLOAD — 02 / EXPLORE THE SPACE

FIND THE OVERLAP ✳ SIGNAL OVER NOISE ✳ FIND THE OVERLAP ✳ SIGNAL OVER NOISE ✳ FIND THE

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MATCHED SKILLS



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MISSING SKILLS



THE MATCH SCORE

71.43%

5.00 of 7 possible skill points · full-vector cosine scoring

5 MATCHED

7 JD SKILLS

LOOK CLOSER / 02

Where you connect.

Matches 5 Missing 2

- Terraform — NO PAIR
- Observability — NO PAIR

VERIFICATION & LIMITATIONS

Functional checks

The repository has 31 Python tests for extraction, scoring, projection, and API errors, plus six JavaScript tests for the map and score display. The local browser flow was exercised with synthetic files and its upload, score, and detail views were captured.

Current limits and next step

The curated vocabulary can miss unfamiliar terms. Image-only PDFs need OCR, and a 3D projection can distort visual distance. An editable preview of extracted skills and an annotated evaluation set would improve review and support measured calibration.

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CONCLUSION

- The app accepts a JD text file and selectable-text resume PDF.
- It scores recognized skills with embeddings and one-to-one matching.
- The result explains the score through points, pairs, and missing JD skills.
- The frontend displays the same result in a rotatable 3D map.
- Human review remains necessary for experience, context, and hiring decisions.